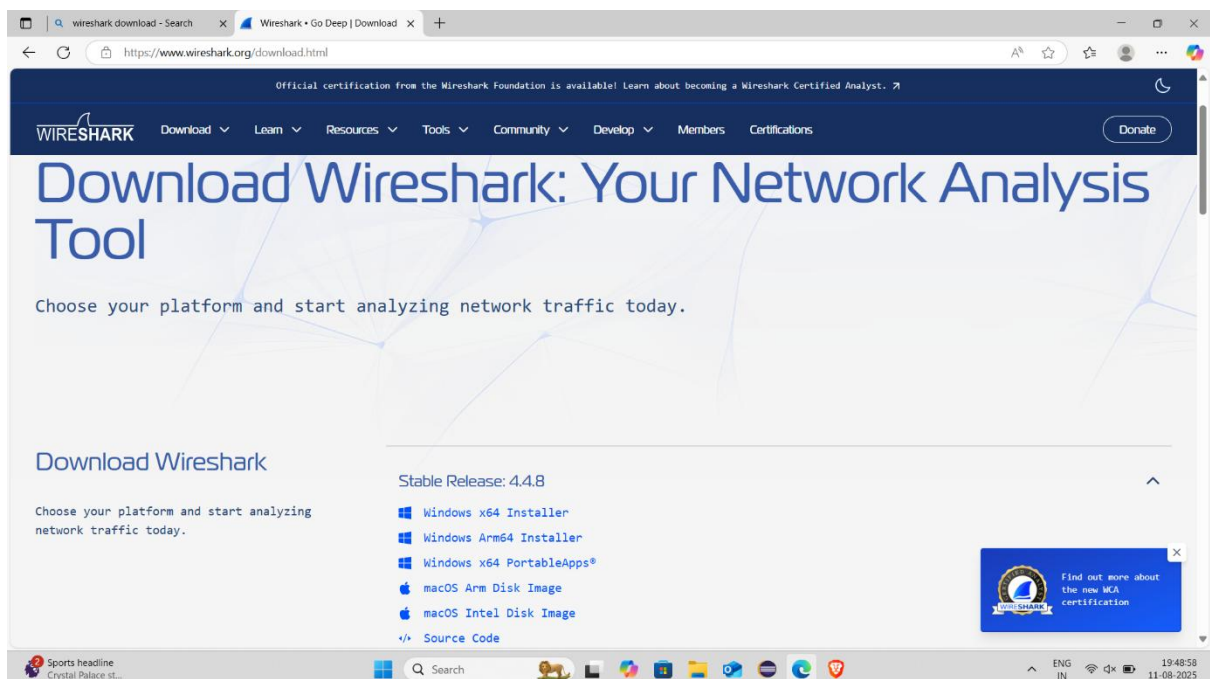


Task 5 : Capture and Analyze Network Traffic Using Wireshark.

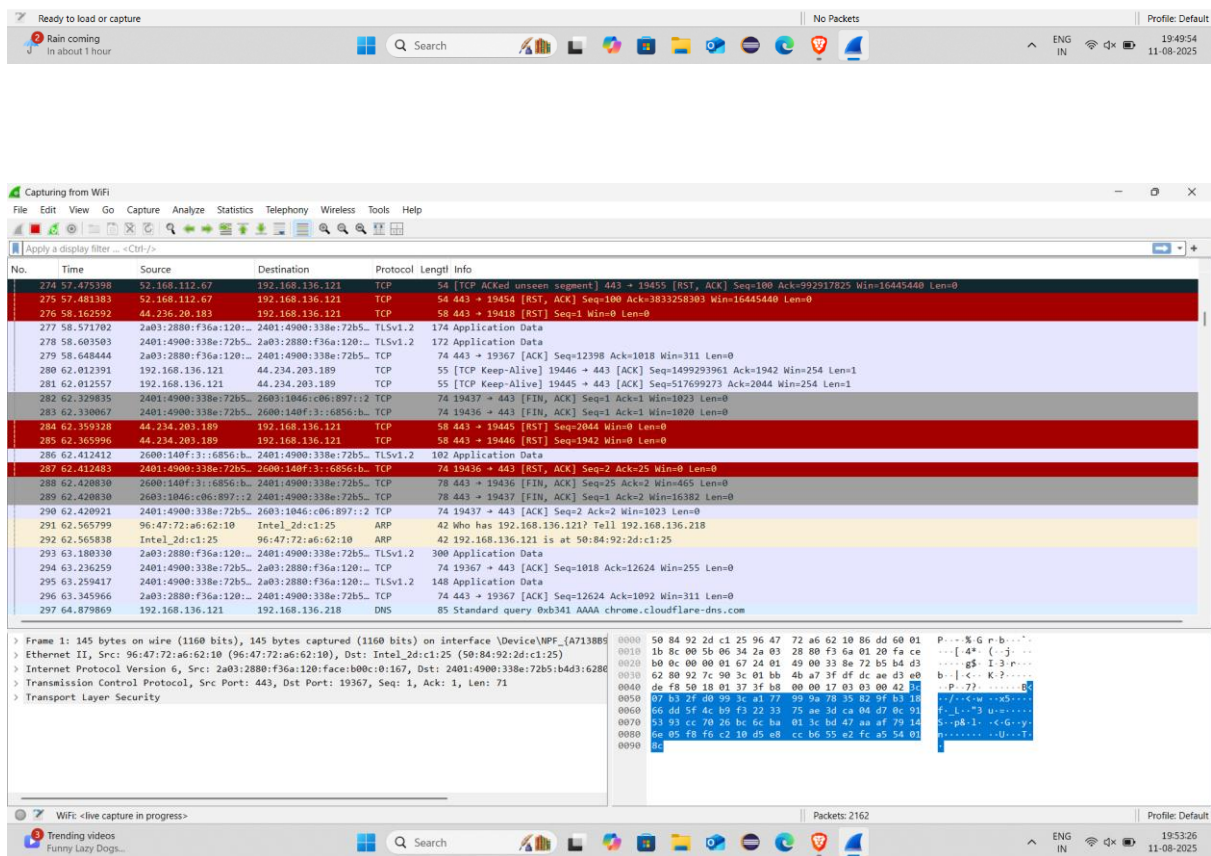
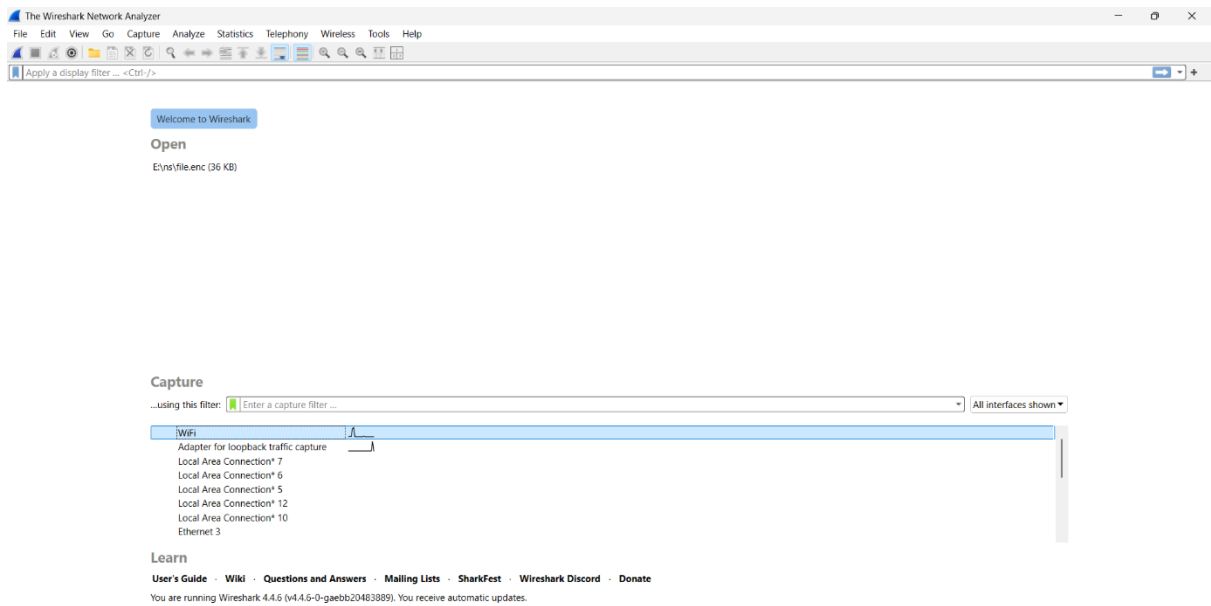
Step 1 – Install Wireshark

1. Go to the Wireshark website.
2. Download the installer for your OS (Windows, Linux, macOS).
3. Install Wireshark with default settings.
 - On Windows, allow installation of **Npcap** (needed for packet capture).



Step 2 – Start a Packet Capture

1. Open Wireshark.
2. You'll see a list of network interfaces (Ethernet, Wi-Fi, etc.).
3. Select your **active network interface** (the one showing live traffic).
4. Click the **blue shark fin** button (Start Capturing).



Step 3 – Generate Network Traffic

To make your capture more interesting:

- Open a browser and visit a few websites.
- You can also send an email, watch a short video, etc.

- Or, in Command Prompt/Terminal, run:

```
C:\WINDOWS\system32\cmd. x + v

Microsoft Windows [Version 10.0.26100.4351]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hari6>ping www.google.com

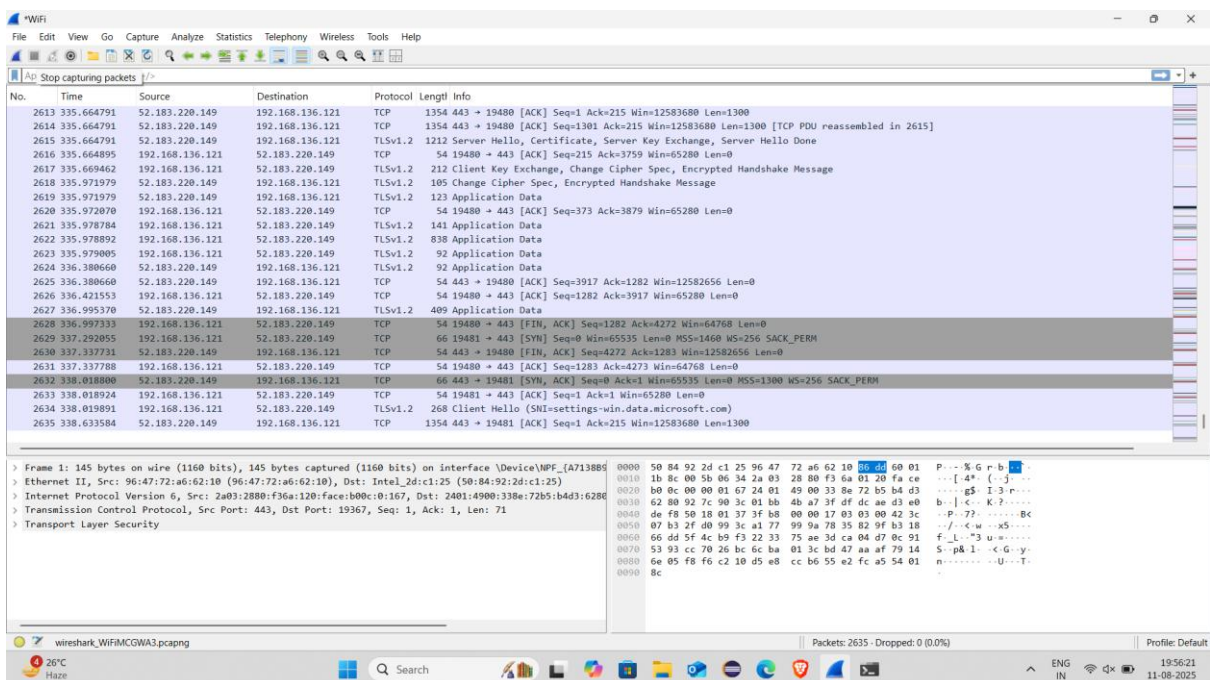
Pinging www.google.com [2404:6800:4007:82e::2004] with 32 bytes of data:
Reply from 2404:6800:4007:82e::2004: time=80ms
Reply from 2404:6800:4007:82e::2004: time=124ms
Reply from 2404:6800:4007:82e::2004: time=93ms
Request timed out.

Ping statistics for 2404:6800:4007:82e::2004:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 80ms, Maximum = 124ms, Average = 99ms

C:\Users\hari6>
```

Step 4 – Stop Capture

1. Let it run for **about 1 minute**.
2. Click the **red square (Stop)** button



Step 5 – Filter by Protocol

Wireshark has a filter bar at the top. Some useful examples:

- HTTP traffic:

The image shows a Wireshark capture of HTTP traffic. The top pane displays a list of packets, with packet 786 selected. The middle pane shows the details of the selected packet, including the Ethernet II header, Internet Protocol Version 6 header, and the Hypertext Transfer Protocol section. The bottom pane shows the raw packet data in hexadecimal and ASCII. The Hypertext Transfer Protocol section indicates a 200 OK response from the server.

No.	Time	Source	Destination	Protocol	Length	Info
783	19.987773	2401:4900:338e:72b5::2	2600:140f:3:1199::2	HTTP	340	GET / HTTP/1.1
786	20.075874	2600:140f:3:1199::2	2401:4900:338e:72b5::2	HTTP	542	HTTP/1.1 200 OK (application/pkix-cert)
797	20.228171	2401:4900:338e:72b5::2	2600:140f:3:1199::2	HTTP	340	GET / HTTP/1.1
802	20.335252	2600:140f:3:1199::2	2401:4900:338e:72b5::2	HTTP	443	HTTP/1.1 200 OK (application/pkix-cert)
856	20.805095	2401:4900:338e:72b5::2	2604:a880:4:1d0::1f	HTTP	485	GET / HTTP/1.1
864	21.123803	2604:a880:4:1d0::1f	2401:4900:338e:72b5::2	HTTP	214	HTTP/1.1 200 OK (text/html)
871	21.197110	2401:4900:338e:72b5::2	2604:a880:4:1d0::1f	HTTP	410	GET /js/init.min.js HTTP/1.1
917	21.538151	2604:a880:4:1d0::1f	2401:4900:338e:72b5::2	HTTP	604	HTTP/1.1 200 OK (application/javascript)
983	21.858405	2401:4900:338e:72b5::2	2604:a880:4:1d0::1f	HTTP	428	GET /css/style.min.css HTTP/1.1
984	21.858577	2401:4900:338e:72b5::2	2604:a880:4:1d0::1f	HTTP	433	GET /css/style-wide.min.css HTTP/1.1
987	22.231690	2604:a880:4:1d0::1f	2401:4900:338e:72b5::2	HTTP	976	HTTP/1.1 200 OK (text/css)
1011	23.045988	2604:a880:4:1d0::1f	2401:4900:338e:72b5::2	HTTP	556	HTTP/1.1 200 OK (text/css)
1093	23.473441	2401:4900:338e:72b5::2	2604:a880:4:1d0::1f	HTTP	495	GET /css/images/banner.svg HTTP/1.1
1095	23.479537	2401:4900:338e:72b5::2	2604:a880:4:1d0::1f	HTTP	510	GET /css/images/header-major-on-light.svg HTTP/1.1
1222	23.799378	2604:a880:4:1d0::1f	2401:4900:338e:72b5::2	HTTP	1335	HTTP/1.1 200 OK
1223	23.801686	2401:4900:338e:72b5::2	2604:a880:4:1d0::1f	HTTP	468	GET /favicon.ico HTTP/1.1
1226	23.810374	2401:4900:338e:72b5::2	2604:a880:4:1d0::1f	HTTP	509	GET /css/images/header-major-on-dark.svg HTTP/1.1
1228	23.820824	2604:a880:4:1d0::1f	2401:4900:338e:72b5::2	HTTP	85	HTTP/1.1 200 OK
1316	24.085400	2604:a880:4:1d0::1f	2401:4900:338e:72b5::2	HTTP	1271	HTTP/1.1 200 OK (image/x-icon)
1319	24.089362	2604:a880:4:1d0::1f	2401:4900:338e:72b5::2	HTTP	1341	HTTP/1.1 200 OK

- DNS traffic

The image shows a Wireshark capture of DNS traffic. The top pane displays a list of packets, with packet 11 selected. The middle pane shows the details of the selected packet, including the Ethernet II header, Internet Protocol Version 4 header, User Datagram Protocol header, and the Domain Name System (query) section. The bottom pane shows the raw packet data in hexadecimal and ASCII. The Domain Name System section indicates a query for the domain 'assets.msn.com'.

No.	Time	Source	Destination	Protocol	Length	Info
12	9.291753	192.168.136.121	192.168.136.218	DNS	88	Standard query 0x4d96 AAAA static.edge.microsoftapp.net
13	9.291913	192.168.136.121	192.168.136.218	DNS	88	Standard query 0x4daf A static.edge.microsoftapp.net
16	9.360808	192.168.136.218	192.168.136.121	DNS	316	Standard query response 0x4daf A static.edge.microsoftapp.net CNAME edge-cloud-resource-static.azureedge.net CNAME edge-cloud-resource-static.azureedge.net
19	9.482288	192.168.136.218	192.168.136.121	DNS	328	Standard query response 0x4daf A static.edge.microsoftapp.net CNAME edge-cloud-resource-static.azureedge.net CNAME edge-cloud-resource-static.azureedge.net
20	9.488004	192.168.136.218	192.168.136.121	DNS	360	Standard query response 0x4daf A static.edge.microsoftapp.net CNAME edge-cloud-resource-static.azureedge.net CNAME edge-cloud-resource-static.azureedge.net
108	24.199132	192.168.136.121	192.168.136.218	DNS	91	Standard query 0x0592 AAAA teams.events.data.microsoft.com
109	24.199278	192.168.136.121	192.168.136.218	DNS	91	Standard query 0x059a A teams.events.data.microsoft.com
110	24.199560	192.168.136.121	192.168.136.218	DNS	91	Standard query 0x0605 HTTPS teams.events.data.microsoft.com
111	24.207300	192.168.136.218	192.168.136.121	DNS	210	Standard query response 0x059a A teams.events.data.microsoft.com CNAME teams-events-data.trafficmanager.net CNAME onedcolprdeus
112	24.293944	192.168.136.218	192.168.136.121	DNS	255	Standard query response 0x0605 HTTPS teams.events.data.microsoft.com CNAME teams-events-data.trafficmanager.net CNAME onedcolprdeus
113	24.293944	192.168.136.218	192.168.136.121	DNS	255	Standard query response 0x0592 AAAA teams.events.data.microsoft.com CNAME teams-events-data.trafficmanager.net CNAME onedcolprdeus
171	27.609463	192.168.136.121	192.168.136.218	DNS	74	Standard query 0x4dd9 A assets.msn.com
172	27.609865	192.168.136.121	192.168.136.218	DNS	74	Standard query 0xe321 AAAA assets.msn.com
173	27.691108	192.168.136.218	192.168.136.121	DNS	246	Standard query response 0x4dd9 A assets.msn.com CNAME assets.msn-com-world-atm-default.trafficmanager.net CNAME assets.msn-com-world-atm-default.trafficmanager.net
174	27.702255	192.168.136.218	192.168.136.121	DNS	270	Standard query response 0xe321 AAAA assets.msn.com CNAME assets.msn-com-world-atm-default.trafficmanager.net CNAME assets.msn-com-world-atm-default.trafficmanager.net
297	64.879869	192.168.136.121	192.168.136.218	DNS	85	Standard query 0xb341 AAAA chrome.cloudflare-dns.com
298	64.880308	192.168.136.121	192.168.136.218	DNS	85	Standard query 0xf6d4 A chrome.cloudflare-dns.com
299	64.880675	192.168.136.121	192.168.136.218	DNS	85	Standard query 0xf6d4 HTTPS chrome.cloudflare-dns.com
300	64.885999	192.168.136.218	192.168.136.121	DNS	141	Standard query response 0xb341 AAAA chrome.cloudflare-dns.com AAAA 2a06:98c1:52::3 AAAA 2803:f800:53::3
301	64.886993	192.168.136.218	192.168.136.121	DNS	117	Standard query response 0xf6d4 A chrome.cloudflare-dns.com A 172.64.41.3 A 162.159.61.3
308	64.978001	192.168.136.218	192.168.136.121	DNS	158	Standard query response 0xf6d4 HTTPS chrome.cloudflare-dns.com HTTPS
391	73.444577	192.168.136.121	192.168.136.218	DNS	86	Standard query 0x7dd6 A slscr.update.microsoft.com
392	73.444860	192.168.136.121	192.168.136.218	DNS	86	Standard query 0xb024 AAAA slscr.update.microsoft.com

- TCP traffic:

The image shows a Wireshark capture of network traffic. The top pane displays a list of packets, with the 'tcp' filter applied. The middle pane shows the details of the selected packet (No. 322), including the Ethernet II, Internet Protocol Version 6, and Transmission Control Protocol (TCP) layers. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
322	64.994681	192.168.136.121	52.24.121.233	TCP	66	19456 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

Frame 1: 145 bytes on wire (1160 bits), 145 bytes captured (1160 bits) on interface \Device\NPF... (A7138B8...)

Ethernet II, Src: 96:47:72:a6:62:10 (96:47:72:a6:62:10), Dst: Intel_2d:c1:25 (50:84:92:2d:c1:25)

Internet Protocol Version 6, Src: 2a03:2880:f36a:120:face:b00c:0:167, Dst: 2a01:4900:338e:72b5:b4d3:628e

Transmission Control Protocol, Src Port: 443, Dst Port: 19367, Seq: 1, Ack: 1, Len: 71

Transport Layer Security

TLSv1.2 Record Layer: Application Data Protocol: Hypertext Transfer Protocol

Content Type: Application Data (23)

Version: TLS 1.2 (0x0303)

Length: 66

Encrypted Application Data: 3c07b32fd0993ca177999a7835829fb31866d45f4cb9f3223375ae3dca04d70c915393

Application Data Protocol: Hypertext Transfer Protocol

- Others(TLS & SSDP)

The image shows a Wireshark capture of network traffic. The top pane displays a list of packets, with the 'ssdp' filter applied. The middle pane shows the details of the selected packet (No. 759), including the Ethernet II, Internet Protocol Version 4, and User Datagram Protocol (UDP) layers. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
759	130.499889	fe80::e622:a652:687...ff02::c	ff02::c	SSDP	511	NOTIFY * HTTP/1.1

Frame 11: 88 bytes on wire (704 bits), 88 bytes captured (704 bits) on interface \Device\NPF... (A7138B89D...)

Ethernet II, Src: Intel_2d:c1:25 (50:84:92:2d:c1:25), Dst: 96:47:72:a6:62:10 (96:47:72:a6:62:10)

Internet Protocol Version 4, Src: 192.168.136.121, Dst: 192.168.136.218

User Datagram Protocol, Src Port: 50287, Dst Port: 53

Domain Name System (query)

Wireshark capture showing a TLSv1.2 connection. The packet list shows the handshake process, including Client Hello, Server Hello, Certificate, Server Key Exchange, Server Hello Done, Client Key Exchange, Change Cipher Spec, and Encrypted Handshake Message. The packet details pane shows the TLSv1.2 Record Layer, Application Data Protocol: Hypertext Transfer Protocol, and the encrypted application data. The packet bytes pane shows the raw data in hexadecimal and ASCII.

Step 6 – Identify at Least 3 Protocols

Look for:

- **HTTP** → web browsing data.

Wireshark capture showing an HTTP connection. The packet list shows the GET request and the 200 OK response. The packet details pane shows the HTTP request and response, including the status line, headers, and the body. The packet bytes pane shows the raw data in hexadecimal and ASCII.

- DNS → domain name lookups.

The screenshot shows a Wireshark packet capture on a Windows 10 system. The packet list on the left shows a series of network packets. Packet 11 is selected, which is a DNS Standard query (0xdeb6) for the domain static.edge.microsoftapp.net. The packet details pane on the right shows the structure of the DNS query, including the transaction ID, flags, and the question section. The packet bytes pane on the right shows the raw data of the packet.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	2a03:2880:f36a:120::	2401:4900:338e:72b5::	TLSv1.2	145	Application Data
2	0.041870	2401:4900:338e:72b5::	2a03:2880:f36a:120::	TCP	74	19367 → 443 [ACK] Seq=1 Ack=72 Win=255 Len=0
3	1.945118	44.234.203.189	192.168.136.121	TLSv1.2	409	Application Data
4	1.945118	44.234.203.189	192.168.136.121	TCP	54	443 → 19446 [RST, ACK] Seq=1 Ack=1 Win=64240 Len=0
5	1.945915	44.234.203.189	192.168.136.121	TCP	54	443 → 19445 [RST, ACK] Seq=1 Ack=1 Win=64240 Len=0
6	1.985382	192.168.136.121	4.153.25.42	TCP	54	19450 → 443 [ACK] Seq=1 Ack=356 Win=253 Len=0
7	3.513368	2401:4900:338e:72b5::	2803:f800:53::3	TCP	75	19273 → 443 [ACK] Seq=1 Ack=1 Win=251 Len=1 [TCP PDU reassembled in 2201]
8	3.686344	2803:f800:53::3	2401:4900:338e:72b5::	TCP	90	443 → 19273 [ACK] Seq=1 Ack=2 Win=17 Len=0 SLE=1 SRE=2
9	5.791524	2401:4900:338e:72b5::	2606:4700:6812:f57::	TCP	75	19415 → 443 [ACK] Seq=1 Ack=1 Win=252 Len=1
10	5.891480	2606:4700:6812:f57::	2401:4900:338e:72b5::	TCP	90	443 → 19415 [ACK] Seq=1 Ack=2 Win=17 Len=0 SLE=1 SRE=2
11	9.291582	192.168.136.121	192.168.136.218	DNS	88	Standard query 0xdeb6 AAAA static.edge.microsoftapp.net
12	9.291753	192.168.136.121	192.168.136.218	DNS	88	Standard query 0xdaf A static.edge.microsoftapp.net
13	9.291913	192.168.136.121	192.168.136.218	DNS	88	Standard query 0xalc3 HTTPS static.edge.microsoftapp.net
14	9.294059	2a03:2880:f36a:120::	2401:4900:338e:72b5::	TLSv1.2	161	Application Data
15	9.315400	2401:4900:338e:72b5::	2a03:2880:f36a:120::	TLSv1.2	159	Application Data

Frame 11: 88 bytes on wire (704 bits), 88 bytes captured (704 bits) on interface \Device\NPF_{A7138B91-63...} Ethernet II, Src: Intel_2d:c1:25 (50:84:92:2d:c1:25), Dst: 96:47:72:a6:62:10 (96:47:72:a6:62:10) Internet Protocol Version 4, Src: 192.168.136.121, Dst: 192.168.136.218

User Datagram Protocol, Src Port: 50287, Dst Port: 53

Source Port: 50287
Destination Port: 53
Length: 54
Checksum: 0x92ec [unverified]
[Checksum Status: Unverified]
[Stream index: 0]
[Stream Packet Number: 1]
[Timestamps]
UDP payload (46 bytes)

Domain Name System (Query)

Transaction ID: 0xdeb6
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0

- TCP → transport layer communication.

The screenshot shows a Wireshark packet capture on a Windows 10 system. The packet list on the left shows a series of network packets. Packet 4 is selected, which is a TCP RST, ACK packet. The packet details pane on the right shows the structure of the TCP packet, including the source and destination ports, sequence number, acknowledgment number, and flags. The packet bytes pane on the right shows the raw data of the packet.

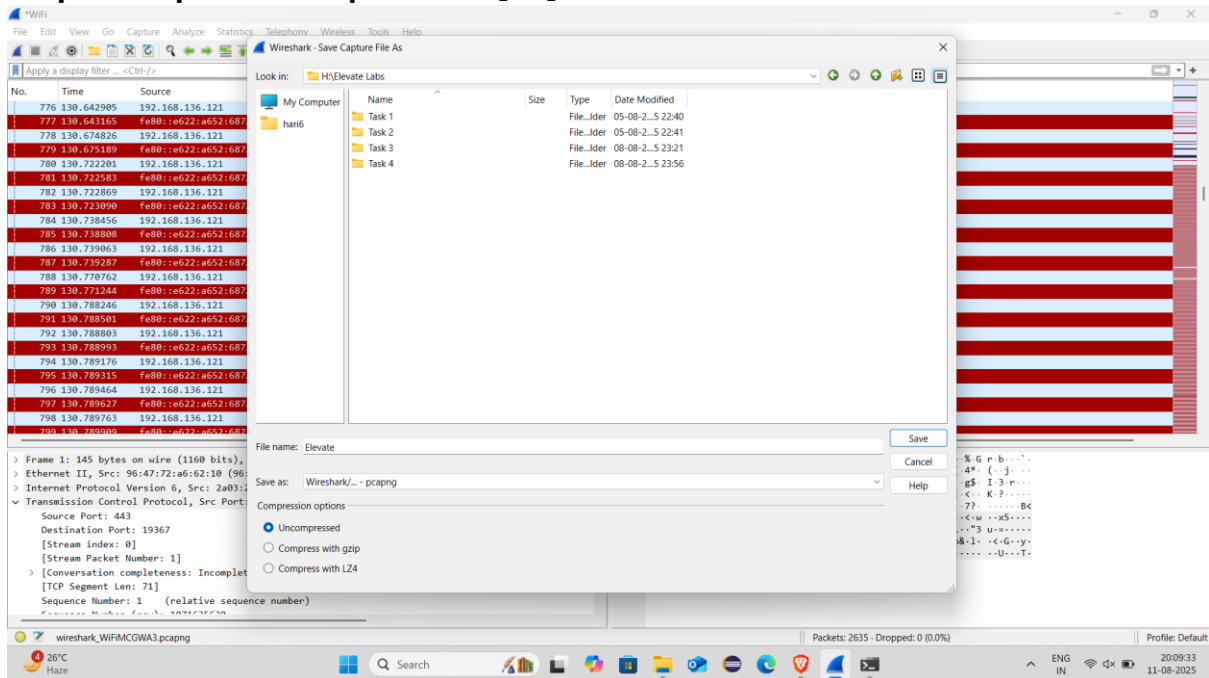
No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	2a03:2880:f36a:120::	2401:4900:338e:72b5::	TLSv1.2	145	Application Data
2	0.041870	2401:4900:338e:72b5::	2a03:2880:f36a:120::	TCP	74	19367 → 443 [ACK] Seq=1 Ack=72 Win=255 Len=0
3	1.945118	44.234.203.189	192.168.136.121	TLSv1.2	409	Application Data
4	1.945118	44.234.203.189	192.168.136.121	TCP	54	443 → 19446 [RST, ACK] Seq=1 Ack=1 Win=64240 Len=0
5	1.945915	44.234.203.189	192.168.136.121	TCP	54	443 → 19445 [RST, ACK] Seq=1 Ack=1 Win=64240 Len=0

Frame 4: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF_{A7138B91-63...} Ethernet II, Src: 96:47:72:a6:62:10 (96:47:72:a6:62:10), Dst: Intel_2d:c1:25 (50:84:92:2d:c1:25) Internet Protocol Version 4, Src: 44.234.203.189, Dst: 192.168.136.121

Transmission Control Protocol, Src Port: 443, Dst Port: 19446, Seq: 1, Ack: 1, Len: 0

Source Port: 443
Destination Port: 19446
[Stream index: 2]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (44)]
[TCP Segment Len: 0]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 4274470105
[Next Sequence Number: 1 (relative ack number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 0
0101 = Header Length: 20 bytes (5)
Flags: 0x0114 (RST, ACK)
Window: 64240
[Calculated window size: 64240]
[Window size scaling factor: -1 (unknown)]
Checksum: 0xe9c3 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]

Step 7 – Export the Capture as .pcap



1. Go to **File** → **Save As**.
2. Choose .pcapng or .pcap format.
3. Name your file and save it for analysis or sharing.